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Quantum computing: a property of matter at nano-scale level that can transform civil engineering to quantum civil engineering

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Abstract
Civil engineering problems are inherently complex and computationally intensive, often requiring advanced methods to analyze large-scale systems and nonlinear interactions. Addressing these challenges requires next-generation computing paradigms that can deliver greater speed and efficiency. Quantum computing, which exploits quantum–mechanical principles such as superposition and entanglement, offers the potential to process information in ways that far surpass classical computing, enabling faster and more accurate solutions to intricate engineering problems. This emerging technology could transform multiple areas within civil engineering—for example, materials science, project management, structural analysis, and design. Continued research and development in this direction may usher the discipline into a new era of computational innovation. Although quantum computing remains at an early stage of maturity, it has already attracted significant attention from researchers across diverse civil-engineering subfields. Fully realizing its potential will require overcoming major technical and practical challenges. This article presents an in-depth review of quantum computing applications in civil engineering, highlighting current progress, advantages, and limitations. It also outlines promising directions for future research to guide the evolution of this rapidly developing interdisciplinary field.

Keywords Quantum computing, Complex problems, Algorithmic advances, Civil infrastructures, Sustainable development

1 Introduction

Quantum mechanics is a branch of physics that describes the behaviour of matter and energy at, and below, the atomic scale [1]. Quantum computing builds upon these principles by harnessing the unique properties of quantum systems—such as superposition and entanglement—to perform computations at the nanoscale [2]. These methods enable extraordinary processing power by exploiting the probabilistic nature of quantum phenomena [2].

Historically, quantum mechanics has been associated mainly with the pure sciences, including biology [3], chemistry [4], and mathematics [5], while its application to

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engineering disciplines has received comparatively limited attention. The abstract nature of quantum theory has often contributed to the perception that it is inaccessible to applied fields. Nevertheless, pedagogical initiatives, such as the visual model by Schalkers et al. [6], which illustrates Grover's search algorithm through an ant-colony analogy, have helped make its principles more comprehensible. Importantly, quantum computation integrates concepts from physics, mathematics, and computer science to address complex computational problems at unprecedented speed.

Modern quantum computing began in 1994, when mathematician Peter Shor introduced a quantum algorithm capable of factoring large integers beyond the reach of classical algorithms. The twenty-first century has since witnessed rapid progress in this domain. Although the field remains in its early stages, it has already influenced diverse disciplines, including biochemical systems [7], drug discovery [8], robotics [9], telecommunications [10], healthcare [11], finance [12], and power systems [13]. Evidence across these areas suggests that quantum computation and simulation can accelerate technological development to meet increasing global demands. Consequently, quantum computing represents a promising avenue for addressing some of the most complex challenges in science and engineering [14].

According to a report by Market.us [15], the global quantum computing market is projected to grow from USD 1.238 billion in 2025 to approximately USD 8.286 billion by 2032, representing a compound annual growth rate (CAGR) of 31.2%, as illustrated in Fig. 1. These figures underscore the rapid expansion and increasing demand for quantum-computing technologies and services worldwide.

One of the key factors driving the growth of the quantum computing market is progress in error correction and fault tolerance technologies [15]. The continued expansion of this market is expected to promote wider acceptance of quantum computing across diverse sectors and industries. As investment and commercialization increase, this growth will likely incentivize scientists, researchers, and innovators to develop and test practical implementations of quantum systems.

Real-world applications are already emerging in areas such as cybersecurity, transportation and traffic management, optimization, and simulation [16]. Figure 2 illustrates

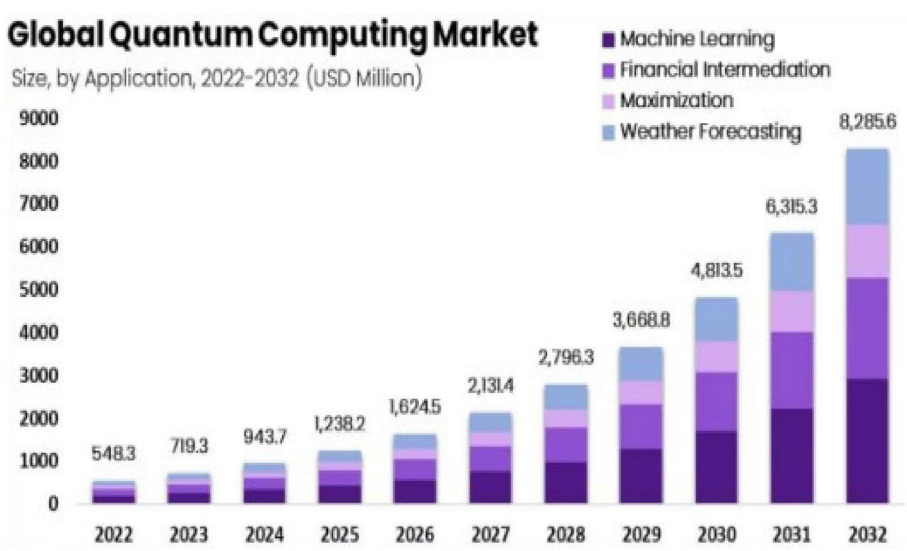

Fig. 1 Global quantum computing market[15]

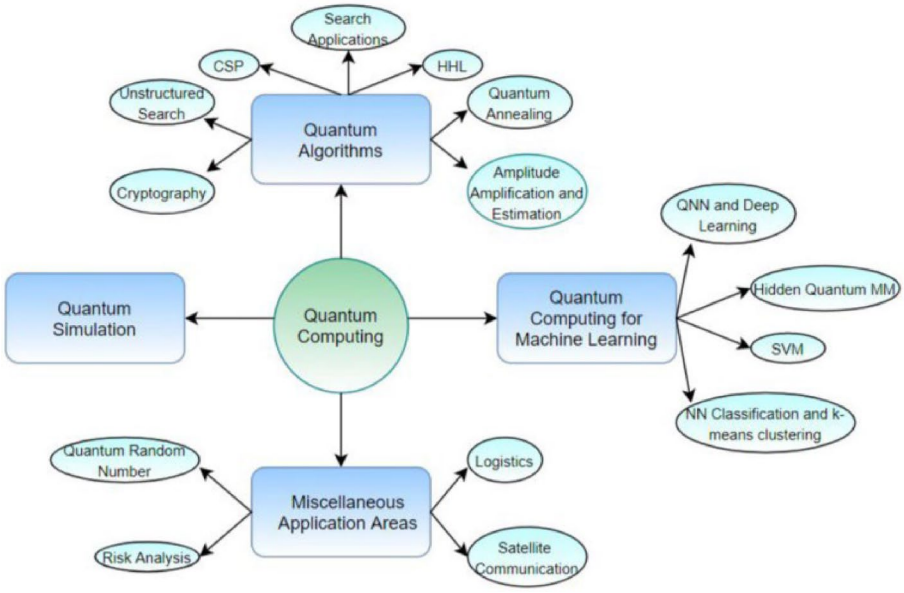

Fig. 2 Quantum algorithm: applications [16]

additional potential applications that could offer scalable solutions to a wide range of real-world engineering and societal challenges.

Sustainability is a central focus in contemporary engineering and policy discourse. Technologies that can protect and monitor ecosystems are essential to achieving long-term environmental goals. Quantum technology offers the potential to drive transformative changes toward a more sustainable future [17, 18]. Its adoption across industries could introduce new levels of speed, precision, and analytical depth in addressing sustainability challenges. Moreover, sustainable quantum-computing technologies may reduce environmental impacts by improving computational efficiency and lowering energy consumption [19, 20]. Applications include accurate prediction of climate patterns and the secure management of data behind green infrastructure systems. For example, in China, a quantum-encrypted network has been established between Beijing and Shanghai to protect data associated with sustainability-related projects, ensuring robustness and security in the future [21].

Engineering optimization problems are inherently nonlinear and often difficult to solve optimally. In civil engineering, structural optimization involves model development, analysis, problem formulation, and computational design platforms [22]. Quantum computing is particularly well suited for such tasks, including optimization, simulation, and complex data analysis. Recent studies have recognized the growing importance of quantum methods in civil-engineering contexts [23–26]. Their potential impact could be profound—ranging from revolutionizing material selection and project management to optimizing construction timelines and traffic systems. Quantum computing thus promises to unlock new levels of innovation and efficiency that can reshape the future of civil infrastructure.

Although quantum computing is not a new concept, only recently has technological progress begun to bridge the gap between theory and practical implementation. As quantum technologies advance, an increasing number of civil engineers capable of collaborating with physicists and computer scientists will be essential. The main aim of this

article is to present an accessible overview of quantum computing for civil engineers and researchers who are not yet familiar with the field, emphasizing its relevance, advantages, and possible research directions. While major real-world breakthroughs may still be several years away, early exploration and interdisciplinary collaboration are crucial to prepare the civil-engineering profession for the quantum era.

2 An overview on quantum computing

The role of quantum computation in complex problem-solving is grounded in the principles of quantum mechanics and computational theory. The fundamental elements of quantum computing—qubits, superposition, and entanglement—enable information to be represented and processed in ways that transcend the limitations of classical computation. These quantum phenomena offer a glimpse into a new era of computation that promises exponential improvements in processing power and efficiency.

2.1 Quantum bit

The fundamental unit of quantum information is the Quantum Bit (Qubit). A qubit, short for quantum bit, can exist in a superposition of basis states, meaning it represents both 0 and 1 simultaneously. This unique property allows quantum computers to perform many calculations in parallel, dramatically enhancing computational capacity [27]

A qubit in a linear state is denoted as [28]

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{1}$$

With α and β as complex numbers in, and $|\alpha|^2 + |\beta|^2 = 1$

A complex unit vector in a two-dimensional Hilbert space with a global state factor is written as[29]

$$|\Psi\rangle = \cos\left(\frac{\theta}{2}\right)|0 + e^{-i\varphi}\sin\left(\frac{\theta}{2}\right)|1\rangle \tag{2}$$

The mentioned vector is described as a point on the exterior of a 3-dimensional sphere, termed the Bloch sphere, shown in Fig. 3.

Mathematically, a general n-qubit quantum state is[6]:

$$|\Psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i|i\rangle \tag{3}$$

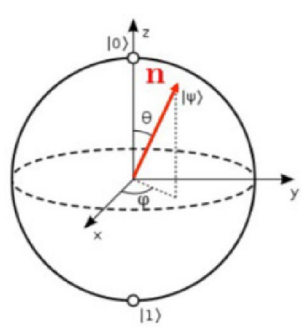

Fig. 3 A qubit state on the Bloch sphere[29]

Quantum algorithms derive their remarkable capacity from intrinsically quantum phenomenon that are considered in the preceding sections. Property such as quantum entanglement and superposition reveals the peculiarities of quantum mechanics.

2.2 Superposition

Superposition is the basic principle of quantum mechanics; it is the ability of qubits to exist in diverse states concurrently [27]. A pure quantum state $|\Psi\rangle$ is a superposition of classical states denoted as [30]

$$|\Psi\rangle = \alpha_0 \,|0 + \alpha_1 \,|1 + \cdots + \alpha_{N-1} \,|N-1 \tag{4}$$

Where α_i is a complex number called the amplitude of $|i\,\rangle$ in $|\Psi\rangle$. Intrinsically, a system in a quantum state is in all classical states simultaneously, each having a specific amplitude. Due to this action, quantum computers can execute numerous computations at the same time, offering exponential speedup over classical computers.

2.3 Entanglement

The concept of how entanglement works has not been fully understood by scientists [31]. Qubits do not react in isolation but become entangled to act as a group [32]. Given qubits A and B in an entangled state is denoted as [33]

$$|\Psi\rangle = \frac{1}{\sqrt{2}}\,[|0_A 0_B\rangle + [\,|1_A 1_B\rangle] \tag{5}$$

Where $|\Psi\,\rangle \neq |\Psi_A\,\rangle \otimes |\Psi_B\,\rangle$

Entanglement is used in quantum parallelism, a capability for quantum computers to perform numerous calculations simultaneously.

2.4 Quantum algorithms

Quantum algorithms are specifically designed to exploit the distinctive properties of quantum computing. These include Shor's, Grover's, Simon's, Bernstein–Vazirani, and Deutsch–Jozsa algorithms, which provide either quadratic or exponential speed-up for specific computational tasks. Among them, the most well-known and experimentally demonstrated are Shor's [34] and Grover's [35] algorithms. These quantum approaches utilize interference to amplify correct solutions and suppress incorrect ones, thereby accelerating the overall problem-solving process.

2.4.1 Grover's algorithm

Grover's algorithm provides a quadratic speed-up for unstructured search problems compared with classical algorithms. To locate a specific element in a database of size N, the quantum solution achieves a time complexity as described in [33]

$$O(N^{\frac{1}{2}}) \tag{6}$$

This quantum technique identifies solutions to search-related problems through the repeated application of a quantum subroutine known as the Grover iteration [27]. Each iteration amplifies the probability amplitude of the desired state while diminishing that of the non-solution states, progressively increasing the likelihood of obtaining the correct result upon measurement [30].

$$\mathcal{G} = H^{\otimes n} R_o H^{\otimes n} O_{x,\pm} \tag{7}$$

Noting that one Grover iterate makes one query, and utilises O(log N) other gates.

2.4.1.1 Grovers operator Grover operator has two quantum operations, namely, mean reflection and marked state inversion [36]. Details of how Grover's algorithm works is stated as follows with Fig. 4 for illustration.

2.4.1.2 Initial state The algorithm starts in a superposition state with all N elements, the state is written as [36]:

$$|\Psi\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle \tag{8}$$

where $|x\rangle$ is the state in accordance to element x.

2.4.1.3 Diffusion operator In quantum operation, diffusion operator multiplies the amplitude of the states corresponded to stated element, it can be denoted as [36]

$$U = \frac{1}{2} I - \frac{1}{2} |\Psi\rangle\langle\Psi| \tag{9}$$

where *I* is the identity operator.

2.4.1.4 Measurement At this stage, the algorithm measures the quantum state, collapsing the superposition to reveal the marked element. Repeated application of the Grover operator amplifies the probability amplitude of the target state, thereby increasing the likelihood of measuring the correct result. Once the target state is reached, the algorithm outputs the corresponding index associated with that state.

In the context of quantum measurement, entanglement refers to the correlation between particles such that the measurement of one influences the state of the other; when no such effect occurs, the particles are not entangled [36].

Grover's algorithm has been implemented in various forms and can enhance classical algorithms that incorporate search procedures by serving as a quantum subroutine [30]. Applications include computational problems such as determining minimum spanning trees, executing graph algorithms, and finding shortest paths.

2.4.2 Shor's algorithm

Factoring large composite numbers is a major challenge in mathematics and forms the basis of many modern cryptographic systems. The development of Shor's factoring algorithm represents one of the most significant milestones in quantum computation [37]. Its prominence arises from the fact that advances in quantum technology enable this

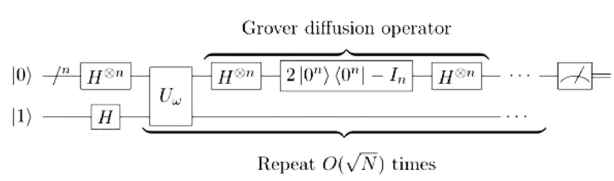

Fig. 4 Grover's algorithm, with k [29]

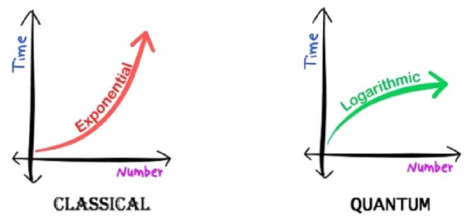

Fig. 5 A graph comparing classical and quantum computation based on time versus number [37]

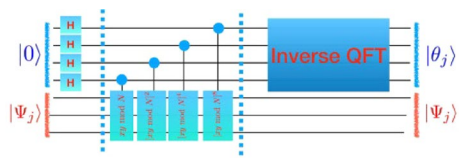

Fig. 6 Shor's algorithm[38]

algorithm to break classical encryption schemes by efficiently decomposing large integers into their prime factors. Figure 5 illustrates the time required to factorize a number using classical and quantum computers.

This algorithm has significant implications for cryptography, as it efficiently factors large composite integers. In the prime factorization problem, the objective is to determine the prime factors of a given integer N using Shor's algorithm. The quantum solution achieves a time complexity as described in [33]

$$O[In(N)^2] \tag{10}$$

Shor's algorithm locates the period r of a function f, given a black box that maps at $|a\rangle|0^n\rangle \mapsto |a\rangle|f(a)\rangle$[30]. Shor's period-finding algorithm is shown in Fig. 6.

Shor's algorithm for factorization is quicker than the conventional algorithm. It can be used to hack RSA and other secure data formations.

2.5 Quantum computing challenges

Quantum computing is a rapidly advancing field that offers numerous advantages. Yet, it faces significant scientific, engineering and practical challenges, as illustrated in Fig. 7. Quantum hardware is still experimental, requires extreme low temperature and is prone to errors and decoherence. In addition, limited practical algorithms, noise resilience, implementation complexity, high cost and manufacturing issues are included.

Despite, the setback researchers are working on mitigation strategies such as better qubit materials, photonic interconnects, noise tolerant and hybrid algorithms. Similarly, efficient QEC codes, logical and topological qubits, cloud access and cryogenic electronics are ways to improve quantum computing.

2.5.1 Decoherence

Decoherence refers to the sensitivity of qubits to their surrounding environment; these interactions cause a gradual decay of quantum behaviour until it eventually disappears [39]. In mesoscopic systems such as qubit devices, decoherence poses a major obstacle to maintaining quantum coherence. However, techniques such as error correction, environmental shielding, and dynamical decoupling are employed to enhance

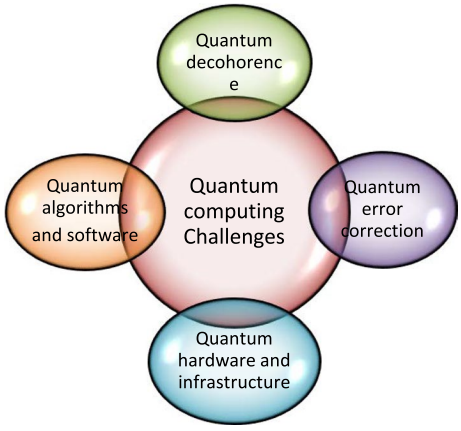

Fig. 7 Challenges of quantum computing

the performance of quantum technologies and prolong coherence times. Furthermore, progress in mitigating decoherence has also emerged from the algorithmic domain rather than purely physical approaches, particularly through the advancement of quantum error correction methods [1].

2.5.2 Error correction

The necessity for error correction is higher for quantum computing than for classical computation. Error correction in quantum world is more difficult for some reasons [30]

1. There no-cloning theorem.
2. It has continuous or infinitely several different possible errors.
3. Measurement test of a correct state can collapse the state resulting to loss of information.

Quantum error-correcting codes encode a single logical qubit into a larger number of physical qubits, allowing errors affecting some of them to be detected and corrected. Fault-tolerant computation represents another essential strategy, designed to prevent the propagation of errors during quantum operations and to ensure the reliability of large-scale quantum systems.

2.5.3 Algorithm and software

Classical probabilistic algorithms can often be simulated efficiently using quantum algorithms. However, the reverse is generally not true—classical probabilistic methods cannot efficiently simulate quantum algorithms. Replacing quantum gates with probabilistic classical gates yields different outcomes and typically fails to reproduce genuine quantum behavior [29].

Many quantum algorithms leverage the unique properties of quantum computation to achieve superior performance. Nevertheless, there remains an urgent need to design and develop new quantum algorithms, a process that requires a deeper understanding of quantum mechanics and its translation into practical algorithmic frameworks. Similarly, developing quantum programming languages presents an ongoing challenge, as the programming paradigms and logic structures underlying quantum computation differ fundamentally from those used in classical computing.

2.5.4 Hardware and infrastructures

Quantum computers are more susceptible to errors than classical digital computers because quantum–mechanical systems are inherently sensitive and difficult to control [29]. Thus, scalability remains one of the major challenges in quantum computing. Building large-scale quantum computers with a sufficient number of qubits continues to pose a significant technical obstacle [14]. The performance of these systems is constrained by factors such as qubit connectivity, coherence time, and gate fidelity. Understanding and addressing these challenges is imperative for the continued advancement and practical implementation of quantum-computing technologies.

2.6 Hybrid quantum computing

Although there are significant challenges as the quantum computing landscape evolves, a wide range of intertwined possibilities would be established that would lead to a future of limitless possibilities. Harnessing classical and quantum computing into a hybrid system can create interesting quantum algorithms, such as a hybrid intelligent quantum algorithm [40, 41]. As an example, quantum computing provides an efficient improvement to Prim's algorithm [42]. Hybrid quantum neural networks enable advantages in optimisation and simulation fields [43]. When quantum convolutions are carried out, the results obtained are fed into a classical neural network to achieve a quantum–classical hybrid model [44]. Hybrid CNN performance converges faster than the quantum version, as Fig. 8 illustrates.

Similarly, Ganesan et al. [45] synergise a combination of quantum computing, evolutionary algorithm, meta-heuristic and multi-objective techniques to create quantum-inspired meta-heuristic(QIMH). The approach solves hard, comprehensive problems to obtain excellent optimisation. Indeed, hybrid algorithms would provide computational speed and a quality solution. This also mitigates some challenges associated with quantum computing.

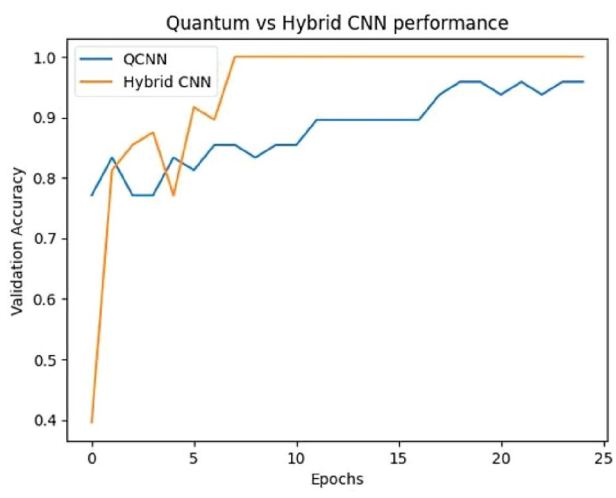

Fig. 8 QCNN versus Hybrid CNN[44]

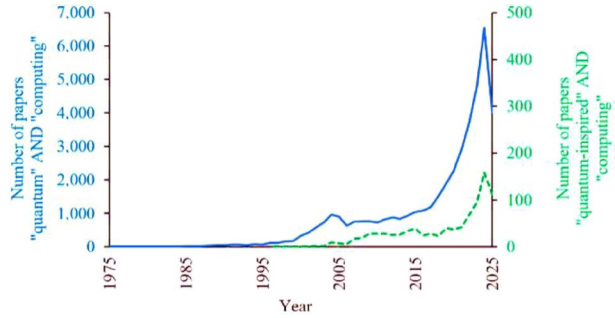

Fig. 9 Published articles on quantum-inspired and quantum computing [46]

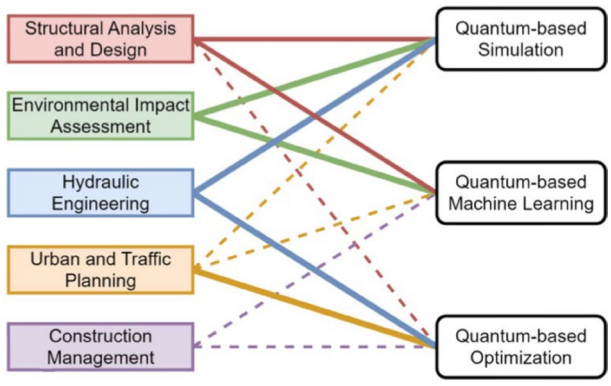

Fig. 10 Mapping of civil engineering subfields with quantum computing potential applicability [24]. Solid lines represent good applicability, dashed line represent some applicability

3 Literature on quantum computing and civil engineering

This section provides insight on current state of research and development in civil engineering with potential cases of quantum computing. Ploennings et al. [46] conducted a multivocal literature review on quantum and quantum-inspired computing in civil engineering.

The result of the bibliography analysis showed a high percentage growth in 2024, illustrating quantum era is arriving. (Fig. 9). Ploennings et al. [46] categories the search string into two SLR (article title, abstract and keywords) and GLR (whole document). Publication output based on SLR for quantum computing related to civil engineering was 23 while quantum-inspired computing was 1. Studies of quantum computing in civil engineering based on GLR was 16 studies while only one was related to civil engineering based on quantum inspired-computing.

Similarly, Smarsly et al. [23] and Ploennings et al. [24] provided an in-depth review of quantum-inspired computing applications in civil engineering—a field that has not yet been fully explored. The authors identified several civil engineering disciplines that could particularly benefit from the adoption of quantum computation. Figure 10 highlights the potential applicability of quantum computing within civil engineering: bold lines indicate areas of high applicability, dotted lines represent moderate applicability, and unmarked connections correspond to limited potential.

Potential applicability of quantum computing has extended beyond computational physics. Civil engineering problems are frequently characterized by high dimensional solution spaces, nonlinear governing equations and computationally intensive numerical

simulation. These challenges are evident in four key problem domains namely optimization, simulation, classification and inverse problems. Table 1, illustrates quantum computing concepts, its computational advantages and specific applicability to civil engineering problems.

The application are emerging and often limited by current noisy intermediate-scale quantum(NISQ) hardware, requiring error mitigation. Full advantages may require fault tolerant quantum computers in the future.

However, quantum technology has the potential to revolutionize every aspect of civil infrastructure, from real-time data analysis to advancements in concrete materials science. The complexity of data in civil engineering has increased significantly due to the rapid growth of large-scale projects, and quantum computing has demonstrated powerful capabilities for processing and analyzing such extensive datasets. There is a growing need to optimize civil engineering systems using quantum genetic algorithms to manage complex data and improve overall infrastructure performance.

The following subsections present a structured review of civil engineering disciplines and their corresponding areas of relevance to quantum computing applications.

3.1 Transportation application

Transportation network design is a well-studied problem concerned with the planning and operation of transportation systems. It is widely applied to identify new roadway links and to enhance existing network efficiency. Dixit and Niu [47] utilized quantum computing to address design problems in transportation networks, demonstrating significant computational benefits achieved through quantum annealing. Wang et al. [48] further illustrated how the advantages of quantum computing can be leveraged to improve existing intelligent transportation systems (ITS).

However, Cooper [49] explored the potential applications of quantum computation in transportation modelling and concluded that implementing quantum algorithms may remain challenging due to the space overheads imposed by the requirements of reversible computation.

3.2 Structural application

Most quantum-computing-based metaheuristic algorithms developed to date have primarily focused on applications in computer engineering. However, Lee et al. [50] applied

Table 1 Quantum computing concepts, computational advantages and applicability to civil engineering

Quantum concepts	Computational advantages	Civil engineering
Superposition	Allows qubits to exist in multiple state simultaneously, enabling parallel exploration of vast solution spaces that could be sequential on classic computers	Implemented in algorithms for optimization, parameter estimation, parallel simulation of multiple scenarios, process high dimensional data for classification
Entanglement	Create correlations between qubits, allowing complex interdepencies to be modelled efficiently, potentially leading to better solutions for interconnected problems	Enhance graph based optimization, inverse modelling, multi physics simulation, classification of interconnected data
Quantum gates	Form quantum circuits to implement algorithms, transforming problems into quantum solvable formats with potential polynomial or exponential speedups for specific tasks	Powers algorithms for solving linear systems, design optimization, builds circuits for quantum simulation algorithms

quantum computing within a metaheuristic framework to analyze truss structures, implementing the developed algorithm for the optimal structural design of a building. Honda et al. [51] formulated a Hamiltonian based on elastic strain energy and the positional configuration of a truss system, confirming that deformation analysis could be effectively performed through quantum annealing. Similarly, Kaveh et al. [52] developed a Quantum Teaching–Learning-Based Optimization (QTLBO) algorithm for discrete sizing optimization of skeletal systems.

Liu et al. [53] established an FEM–VQE pipeline (Finite Element Method–Variational Quantum Eigensolver) to leverage quantum computation in addressing problems in solid mechanics and structural engineering. Martin and Droguett [54] proposed a quantum-based combinatorial optimization approach for solving optimal sensor placement challenges in Structural Health Monitoring (SHM) of civil structures. Their study produced high-quality, near-optimal solutions and outlined potential challenges with proposed mitigation strategies.

Recognizing that shear wall design plays a vital role in tall buildings, Talatahari and Rabiei [55] introduced a Quantum Charged System Search (QCSS) algorithm to enhance the convergence ability of the original Charged System Search (CSS) method. The technique accounted for torsional effects, lateral deflection, flexural and shear resistance, and minimum structural costs. Collectively, these formulations demonstrate the potential of quantum computation to provide more efficient and optimized solutions for structural system.

In addition, Torres et al.[56] proposed a novel approach to identify parameters under seismic excitation using classical and quantum inspired algorithms. The method provided a scalable tool for checking civil infrastructure and probably enables digital twins. Meksen [57] gave insight on exploiting quantum computing for non-destructive testing of materials. Wang and Wang [58] addressed the drawbacks of moving morphable component(MMC) by proposing a quatum behaved particle swarm algorithm(QPSO) for structural topology optimization. The proposed approach demonstrated superior ability in optimizing material arrangements and identifies configurations for structures with complex geometric boundary conditions. Veerasamy et al.[59], proposed an Improved Weighed Quantum Particle Swarm Optimization(IWQPSO) alongside deep convolutions neural network(DCNN) and Gazebo robot simulators to detect concrete surface defect.

3.3 Geotechnical application

Soil properties are complex and difficult to establish an effective surface response for traditional slope stability. Xu et al. [60], developed an intelligent optimisation algorithm by combining the forest regression method(RF) and an improved quantum genetic algorithm. The method identified a critical slip surface for a slope. The distinct properties of soil layers and fluctuation of the bearing stratum is a major challenge in the design of pile foundation. Yin et al. [61], established a BP neural network to predict evaluating samples, then proposed a quantum particle swarm optimization(QPSO) algorithm to optimise weights and thresholds of the BP network. The authors compared the results obtained from QPSO-BP with 3 robust models, FWA-BP, PSO-BP and BP. The accuracy of QPSO-BP was high, which showed superiority amongst the models.

Zhao et al. [62] investigated carbonate rocks' hydraulic properties by using an inspired novel approach. The authors proposed a quantum genetic optimisation to derive an optimal pore network model(PNM). The results obtained from the proposed approach were compared with classical methods, which showed better computational efficiency. The authors concluded that the approach was a promising tool to investigate geomaterials hydraulic properties.

3.4 Hydraulic system

The optimal placement of sensors is key to detecting and isolating faults in Water Distribution Networks (WDNs), which represents a large-scale combinatorial optimization problem. Speziali et al. [63] presented a detailed procedure to address this challenge using adiabatic quantum computation.

Meng and Yang [64] proposed a quantum computing framework based on hydrodynamic Schrodinger equation (HSE) to simulate 3D turbulent flows in different engineering applications. The authors concluded that 3D turbulent structures and statistics have a promising feat in CFD implementation.

The mechanical and physical properties of hydraulic structures evolve over time due to complex operational conditions. Zhu et al. [65] implemented a Quantum Genetic Algorithm (QGA) to inversely analyze the partitioned deformation modulus of high arch dams. The authors concluded that the QGA demonstrated excellent global search capability and population diversity. Furthermore, the feasibility and effectiveness of the algorithm were validated through practical engineering applications.

3.5 Construction management

One major benefit of quantum computing in construction management is its ability to automate complex optimization processes. Oke et al. [66] outlined steps toward the implementation of quantum computing for sustainable infrastructure management. The combination of quantum computing and artificial intelligence (AI) can provide exceptional benefits to construction management by improving efficiency, predictive analysis, and sustainability outcomes.

3.6 Environmental

Environmental engineering plays a crucial role in ensuring the health and well-being of society as well as the environment. New strategies and technologies are continually developed to protect the planet and its ecosystems. Ballari et al. [67] investigated the potential of quantum computation for optimizing environmental engineering processes. The authors concluded that Quantum Computing (QC) offers an innovative and high-speed method for solving complex environmental problems.

Similarly, Arora and Kumar [68] emphasised the importance of environmental sustainability through quantum technologies. They proposed a Carbon-Aware Quantum Computing (CQC) framework and called for a new research direction known as sustainable quantum computing, which aims to minimise the environmental footprint of future quantum systems.

3.7 Damage detection

Damage assessment of buildings following an earthquake is often a labour-intensive, time-consuming, and error-prone process when using traditional methods. However, Quantum Neural Networks (QNNs) have been developed to predict seismic damage more efficiently [69]. The Convolutional Neural Network (CNN) is a well-known algorithm for seismic damage prediction, although it can suffer from delays or inaccuracies in damage evaluation.

Bhatta and Dang [70] presented a novel Quantum Convolutional Neural Network (QCNN) technique to detect damage in reinforced concrete buildings after an earthquake. The QCNN model was trained using damaged images of reinforced concrete structures from previous earthquakes. The seismic damage detection accuracy obtained from the QCNN model was compared with results from traditional CNN architectures, demonstrating improved performance.

For large-scale datasets and faster computational speed, Quantum-Enhanced Machine Learning (QML) offers significant advantages over Classical Machine Learning (CML) [71]. Similarly, Liu [72] proposed a Quantum-Behaviour Particle Swarm Optimisation (QPSO) algorithm to enhance the detection of damage in civil structures. The approach combined a wavelet transform threshold clustering algorithm, neural-network adaptive learning, and Monte Carlo simulation, with the inclusion of Support Vector Machine (SVM) regression and random sequence prediction. Experimental results showed that the minimum wavelet detection error reached 2.4%, indicating that the algorithm provides significant potential for civil infrastructure damage detection.

3.8 Corrosion

Corrosion remains a major issue affecting the reliability and longevity of civil steel structures. Researchers have investigated the potential of quantum computation to analyze materials' corrosivity [73–75]. Akrom et al. [76] evaluated the performance of a Quantum Machine Learning (QML) model to predict the corrosion behavior of stainless steel. The authors found that the Quantum Support Vector Classifier (QSVC) outperformed classical models, achieving accuracies of 95.46% and 94.80% for Dataset A and Dataset B, respectively.

Furthermore, the QML technique demonstrated the ability to predict corrosion behavior without the need for experimental testing, significantly reducing both cost and time. The application of QML in studying corrosion inhibition of compounds has also shown superior performance compared with conventional methodologies [77–79]. In addition, advances in ultra-sensitive quantum sensors hold the potential to enable precise assessment of metal quality and facilitate early and more effective corrosion detection.

3.9 Reliability

Reliability analysis is a critical aspect of ensuring the safety and performance of civil structures. Yazdi et al. [80] discussed the application of quantum computation in structural reliability analysis, emphasizing the efficiency of quantum algorithms in optimizing computational tasks and revolutionizing traditional reliability models.

In a related study, He [81] proposed an effective quantum computation-based structural reliability approach utilizing Quantum Amplitude Estimation (QAE). The proposed method was compared with the well-known Monte Carlo simulation technique.

The study demonstrated that the QAE method significantly reduced computational cost from O(N) to O(√N) for estimating a failure probability of O(1/N), highlighting its substantial computational advantage.

3.10 Construction materials

Construction materials form the backbone of civil infrastructure, playing a crucial role in durability and functionality of modern societies. Karandikar et al.[82] presented a hybrid AI-quantum inspired multi objective framework for self-healing concrete. The frame work improved tremendous the performances of healing efficiency, compressive strength and reduced lifecycle cost. Similarly, Panse et al.[83] utilized quantum kernel support vector and multi objective optimization techniques that integrates nano-alumina and zeolites to predict and optimize methodology for high performance concrete(HPC). An efficient framework was established, that optimizes HPC formulations and reduced CO_2 outputs by 20%

3.11 Insights from literature based on quantum computing approaches and experimental implementation in civil engineering

Upon the review, of some quantum computation in civil engineering, most approaches were demonstrated through simulations or hybrid systems rather than fully on fault-tolerant quantum hardware. Table 2 summarizes.

Quantum computing applications in these subfields are at proof of concept stage, but it has demonstrated it versatility across civil engineering disciples. This illustrates its potential to transform and modernize traditional analytical and computational methodologies.

4 Proposed and speculative hybrid algorithms

The application of theorems and algorithms developed within quantum theory over past decades has primarily focused on disciplines outside civil engineering. To achieve greater efficiency in decision-making, resource allocation, and simulation, quantum computing offers the advantage of faster data processing and enhanced algorithmic capability [84]. The exploration of quantum-inspired algorithms to solve fundamental challenges—such as the shortest path problem—promises valuable outcomes for various subfields within

Table 2 Summary of quantum computing approaches and experimental implementation

Quantum computing approach	Experimental/implementation details
Quantum annealing (QA) [47, 51, 63]	Sensor placement, project scheduling, hydrologic inverse analysis, pile foundation optimal design, transportation networks
Quantum teaching–learning-based optimization (QTLBO) [52]	Optimizes skeletal structures
Quantum charged system search (QCSS) [55]	Structural analysis and cost evaluation
Quantum-behavior particle swarm optimization (QPSO) [58, 59, 61, 72]	Damage detection, soil parameters prediction, structural topology
Quantum genetic algorithm (QGA) [63, 65]	Slope stability, Inverse analysis in dams
Carbon-aware quantum computing (CQC) [68]	Carbon footprint, green design
Quantum neural networks(QNN) [69]	Seismic damage
Quantum convolutional neural network (QCNN) [70]	Damage detection, flood prediction
Quantum machine learning (QML) [76]	Corrosion, prediction analysis, structural health monitoring
Quantum amplitude estimation (QAE) [81]	Structural reliability for estimating failure probabilities

civil engineering. The following outlines potential prospects arising from the exceptional efficiency of quantum computation.

1. The Finite Element Analysis (FEA) method is one of the most widely used techniques for detailed structural investigation [85]. FEA can be significantly enhanced by leveraging quantum computing. The Harrow–Hassidim–Lloyd (HHL) algorithm is a well-known quantum algorithm capable of solving linear systems of equations [86], which are fundamental to FEA. Integrating this algorithm could accelerate simulation processes, enable faster design iterations, and improve the precision of structural analyses.

In analyzing engineering systems, selecting appropriate variables to represent system behavior is crucial. For example, forces or deflections characterize structural systems; motions or velocities define mechanisms; and heat fluxes or temperatures describe heat transfer problems. These are often represented through matrix formulations.

A matrix representation of a three-qubit quantum truss can be constructed from governing equations, enabling the calculation of displacement and rotation values of the system. Similarly, this quantum conceptualization could be extended to traffic flow optimization, resource allocation in construction management, and mechanical system computations, significantly improving analytical performance and efficiency.

2. Graph formalism enables eigenvalue conditions to be expressed as secular equations for matrices of finite dimension. However, when an engineering system is not well defined, the resulting mathematical manipulations may lead to inconsistent or unsolvable formulations [87].

Quantum graph theory is an active research area that establishes isomorphisms between quantum states and graph structures, providing a framework for developing quantum-isomorphic representations applicable to civil engineering system analysis. Musto et al. [88] demonstrated that graphs quantum-isomorphic to a given graph are in bijective correspondence with Morita equivalence classes possessing finite-dimensional representations. Similarly, Lawrie et al. [89] identified quantum graph theory as an ideal mathematical framework for modelling engineering metamaterials.

The combination of graph theory with quantum mechanics could lead to novel approaches for analyzing vibrational states in engineering structures. This integration also supports multidisciplinary applications, including vibration modelling of coupled plates, quantum walks, and search algorithms [90]. For instance, quantum minimum-finding algorithms can be adapted to solve problems such as the shortest path and graph connectivity in rail infrastructure networks, leveraging Grover's algorithm for enhanced computational efficiency.

3. Combinatorial Representation (CR) is a foundational concept rooted in discrete mathematics [87]. A Quantum Combinatorial Representation (QCR) can be developed by integrating discrete linear programming, matroid theory, graph theory, and quantum theory. Through this approach, new proofs of established theorems and computational methods can be formulated in novel ways—now expressed within the framework of quantum combinatorial theorems embedded in QCR.

Such representations would open new avenues for developing quantum-based algorithms in machine reasoning and system optimization. In particular, the development of quantum graph-theoretic representations for civil engineering systems could significantly enhance the interpretability and reasoning of complex structural and infrastructural models.

4. Quantum walks represent a fascinating paradigm capable of discovering new classes of quantum algorithms. For example, an optimal quantum algorithm for element disparity problems can be derived using a quantum walk algorithm [29]. A quantum walk at any step t is a unitary transformation applied to the current state $|\Phi^{t-1}\rangle$, expressed as [91]:

$$U_{QW}|\Phi^{t-1}\rangle = |\Phi^{t}\rangle \tag{11}$$

Where t = 0, $|\Phi^{0}\rangle$ symbolises the first quantum state of graph G.

The utilisation of quantum walk algorithm for graph presentation can efficiently encode topological data of graphs into quantum states [92]. This combination would definitely provide unprecedented efficiency in graph connectivity analysis, shortest path finding, and network centrality measures for civil infrastructures. In addition, to solve low computational efficiency of traditional critical path method, an accelerated model based on ant colony optimization [92] with quantum walk and Grover's algorithm for CPM can assist construction personnel to ensure swift schedule and reasonable resource allocations in critical path.

5. Particle swarm optimisation is the most common swarm intelligence algorithm to solve optimisation problems [93]. The hybrid particle swarm optimisation algorithm has proven to be efficient in solving engineering problems, but it takes more computational time [94]. In improving this, applying quantum computation can improve speed. Cable-strut systems rely mainly on prestressing to provide stiffness, making prestressing design for the system crucial [95]. To determine optimal prestress modes for cable-strut systems, a hybrid quantum optimiser is proposed, inspired by salp swarm [96], termite [97, 98] and quantum algorithm (QSSTA). The reaction of initial prestress, internal infinitesimal, can be rigidified to achieve the desired structural stiffness through the implementation of the proposed algorithm.
6. Ladybug beetle optimisation (LBO) algorithm obtained optimal values of 73.3% functional benchmark and the only algorithm to achieve efficiency of 20.5% solution [99]. A state-of-the-art algorithm with the Quantum and ladybug beetle (QLBO) algorithm can be established and used for modelling problems to obtain little or zero errors for flood risk mitigation. A bridge is a crucial construction system, and its optimisation requires extensive documentation. The stated hybrid quantum heuristic search algorithm can be implemented to estimate low cost, number of reinforcing bars and CO_2 emissions in bridge works. Quantum-enhanced vibrating particle system can be developed to optimise the cost of reinforced concrete box-girder. In addition, Zhu et al. [100], proposed a dung-beetle search algorithm(QHDB) based on multi-strategy hybrid and quantum computing to solve engineering problems. The findings illustrated that the studied algorithm improved accuracy, converged speed and has excellent robustness. This algorithm can be integrated with other optimisation to

mitigate civil engineering complex problems such as rail infrastructures, skyscrapers and bridges as earlier stated.

7. The Firefly Algorithm (FA) was inspired by the bioluminescent behavior of fireflies [101]. It offers advantages over other swarm intelligence algorithms in terms of optimization speed and solution accuracy [102]. However, the algorithm suffers from limitations such as premature convergence, a weak theoretical foundation, and sensitivity to parameter selection [103, 104].

Zhang et al. [105] developed a hybrid Firefly–Butterfly Optimization Algorithm to solve a three-bar truss and speed reducer design problem. The proposed hybrid algorithm demonstrated superior performance and outperformed other optimization techniques in terms of stability and accuracy.

The integration of Firefly, Butterfly, and Quantum algorithms could further enhance computational performance, providing faster convergence and improved global search capability. Such quantum–bio-inspired hybrid algorithms may offer powerful solutions for addressing the complex optimization challenges frequently encountered in civil engineering design and analysis.

8. The Grey Wolf Optimization (GWO) algorithm is a metaheuristic inspired by the leadership hierarchy and hunting behavior of grey wolves [106]. It is simple to implement [107], yet it faces several drawbacks, including slow convergence speed, low solution accuracy, and a tendency to fall into local optima [95].

Li et al. [108] proposed an Improved Grey Wolf Optimization (IGWO) algorithm to overcome these challenges in engineering design problems. The IGWO method effectively minimized resource waste and enhanced design efficiency.

Leveraging the strengths of quantum computation, a Quantum-Improved Grey Wolf Optimization (QIGWO) algorithm could be developed to further enhance neuro-fuzzy inference systems and multilayer perceptron models. Such an approach can improve the prediction accuracy of complex engineering parameters, such as the ultimate bearing capacity of driven piles, providing a significant advancement over conventional heuristic optimization methods.

9. Water-structure interaction, soil-structure interaction and soil–water-structure interaction are extremely complex systems because of soil non-linearities, multi-body interface and multi-phase coupling impact. Bio-inspired jellyfish search algorithm has proven to solve optimization problems [109, 110]. The hybridization of quantum algorithm like Grover's and jellyfish algorithm can drastically improve the search for unknown parameters. Estimating active and passive pressure for retaining structures, solving optimal solution with multiple objectives and overcoming drawbacks can be achieved with Quantum Jellyfish Optimization (QJO) algorithm.
10. The Harmony Search Algorithm (HSA) is inspired by the performance of a musical ensemble striving to achieve a harmonious state through the combination of various instrumental pitches [111]. In this analogy, design variables that influence the objective function are compared to instrumental sounds contributing to musical harmony. Owing to its simplicity and flexibility, HSA has been widely implemented across diverse engineering domains to solve complex optimization problems [112].

In a study on the behavioral response of a suspension dome with CFRP cables, the harmonic concept was applied by modeling the system's behavior as analogous to wind-induced wave patterns [113]. Building on this concept, a Quantum Harmony Search Algorithm (QHSA) can be developed to further enhance optimization capability.

The integration of quantum computing principles with the traditional harmony search model could improve global search efficiency and convergence speed. A QHSA could be employed to optimize the design of steel frames with semi-rigid connections, analyze vibrational states of structures, enhance heat transfer design, and improve acoustic localization in structural environments.

11. Understanding the physics of materials requires advanced computational tools capable of simulating behavior at the atomic and molecular levels. To address this, the Material Generation Algorithm (MGA) was developed for solving complex optimization problems in material design [114]. The algorithm models chemical reactions and compound formations to support the creation of new materials inspired by natural processes.

By hybridizing quantum computation with MGA to form a Quantum Material Generation Algorithm (QMGA), researchers can potentially create sustainable construction materials with superior properties such as lightweight strength, self-healing ability, and ultra-high durability.

12. Owing to its inherent ability to simulate quantum systems at the atomic level, quantum computing can revolutionize material simulation and discovery [115]. This approach enables precise prediction of molecular structures, material properties, and chemical behavior, allowing researchers to optimize performance, accelerate material discovery, and tailor novel material designs for specific engineering applications.
13. To ensure probability stability evaluation of soil slopes. Quantum support vector regression (QSVR) [116] can be optimized with a chaotic and neighbourhood search based artificial bee colony algorithm [117] to establish a response surface and approximate limit state function. The effect of rainfall, spatial variability and external loading on slope stability can be investigated through the proposed
14. Krill herd algorithm (KH) is a new optimisation algorithm [118], which is based on simulated forging behaviour by krill individuals. The algorithm satisfies many optimisation problems yet it can get stuck in certain local optima and cannot be applied fully in global search [119]. In terms of global best achievement, the alligator optimisation algorithm (Agtro) outperformed other algorithms [120]. A hybrid krill herd algorithm with alligator and quantum computing can mitigate the setback of the krill herd algorithm. The development of the QKH-Agtro algorithm can be implemented for civil infrastructure optimisations.
15. The rapid development of highway engineering, traffic path planning, and quality monitoring has new challenges; mitigating these problems would require creating new models.in addition, due to the growing demand of civil infrastructures, effective data management is required to improve project management data flexibility. Octopus Octopus-inspired optimisation (OIO) algorithm is efficient for complex optimisation problems [121]. Harnessing quantum computing with OIO algorithm (QOIO), to create a state-of-the-art algorithm for the stated challenges would definitely give exceptional findings.

16. Structural health monitoring predicts durability for engineering structures under severe weather conditions. The development of new quantum-based optimisation techniques for structural health monitoring is essential [122]. Artificial neural networks have gained considerable attention as a surrogate for structural health monitoring [123] . However, the traditional neural network has poor convergence under training of local search optimization algorithm and is not reliable [124, 125]. A hybrid model can be established, a non-linear mapping ability of artificial neural network (ANN) model can be infused with quantum computing principles. Evaluating bond properties of reinforcement bars to concrete is of great importance for the design and application of polymer matrix composites [104]. The proposed algorithmic (QANN-quantum artificial neural network) can offer a solution to accurately predict the bonding strength of bars to concrete.
17. Genetic algorithm (GA) is one of the famous evolutionary algorithms which uses people association that endures the presence of mutability-compelling factors like mutation and crossover[103]. It is simple to perform, however, there is no guarantee that the global maximum will be attained.it has the possibility of been trapped in the local maximum. With a large number of variables, using GA can be time-consuming. With the beneficiary traits of the Grovers algorithm [35], hybridisation of the two algorithms is proposed as $H(GA)^2$. Exploring this hybrid algorithm has the potential ability of detecting damage, since structural health monitoring (SHM) is a hot topic for civil infrastructures.

With respect to the slow convergence rate of the genetic algorithm in large-scale optimisation, quantum computation can leverage this setback. Hybrid optimisation between quantum and genetic algorithms can pave the way for novel algorithms. These algorithms can be applied to various engineering challenges prediction of cracks, welded beam optimisation, truss size optimisation, tubular column design and cantilever beam design.

18. Machine learning (ML) and artificial intelligence (AI) are technologies known to offer alternative means to solve complex and ill-defined problems [104]. It learns from examples, has a powerful fault tolerance, meaning the technology can deal with noisy and incomplete data and non-linear challenges [126]. AI has opened new avenues for automated control of quantum systems [127]. Integrating AI with quantum computing in creating useful applications is still unclear [128]. However, Singh et al. [129] argued that quantum artificial intelligence(QAI) may offer key discoveries required to further climate change science. The establishment of QAI would be beneficiary for water resource engineering, especially for flood mitigation, surface and groundwater hydrology.

Recently, research has established quantum algorithms that act as cornerstones for machine learning programs [130–132]. Quantum machine learning (QML) can drive the development of novel algorithms and frameworks for civil infrastructures. Quantum-Boltzmann machine learning method (QBML) and propagation neural network (QPNN) model can be developed for construction cost estimation. Detecting surface damage like cracks, corrosion, and delamination can be achieved with a quantum hybrid convolutional neural network (QHCNN) without calculating defect characteristics. Prediction

of pavement friction, shear strength and dynamic problems can easily be obtained through QHCNN.

19. Hybrid quantum–classical computing was developed to exploit its merits from the assumed performance accretion provided by quantum technology [133]. Figure 11 illustrates an optimiser cycle between classical and quantum computation. In combinatorial optimisation problems, quantum approximate optimisation algorithm (QAOA) is a recognised variational optimization quantum algorithm for decoding QUBO problems.

Synergy between machine learning and quantum computing has demonstrated stability and adaptability [135].Graph neural network has improved quantum approximate optimization algorithm(QAOA). The implementation of quantum approximate optimization algorithm (QAOA) [136, 137] conversion of issues like traveling salesman problem into quadratic unconstrained binary optimization (QUBO)[138, 139] and graph neural network (GNN)can be implemented in complex situation like uncertainty and variability in civil infrastructure designs, cost-effective construction and earth quake resistant designs.

20. To create a powerful search and design optimizer for structures with discrete-sized member, a quantum inspired approach with mother optimization algorithm (MOA) [140] and crow search algorithm CSA [141] is proposed. Quantum mother optimization-crow search algorithm (QMOCSA) can be implemented for the optimal design of large-scale steel frames and anchorage.
21. A bio-quantum swarm negotiation (BQSN) is proposed, this algorithm integrates bio-inspired swarm intelligence, quantum computing principles and negotiation theory to solve complex, multi-agent optimization problems in dynamic environment.

The above discussions are some future theory for the amalgamation of quantum technology and civil engineering. Difficult civil engineering problems can be solved by developing and hybridizing standard algorithms with quantum computing. Galvao et al. [142], reported the significance of introducing quantum computing teaching practices in high school education, highlighting its scientific as well as industrial relevance. The

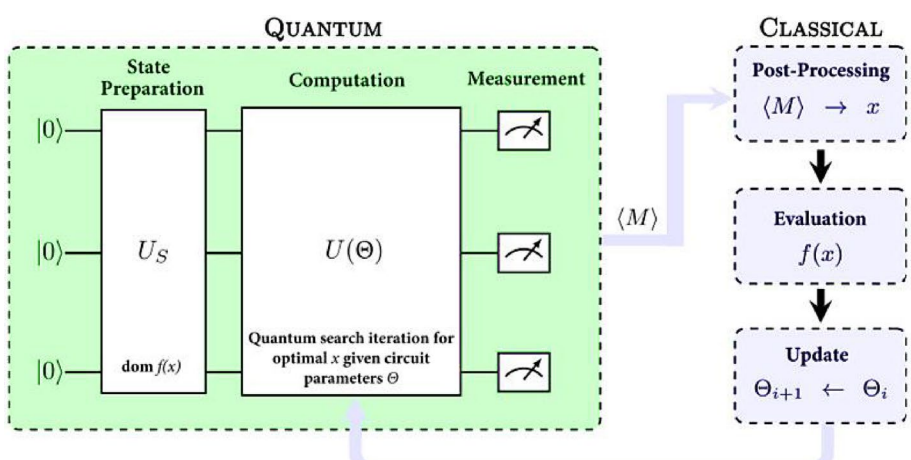

Fig. 11 Hybrid quantum classical optimization workflow [134]

implementation of quantum physics in civil engineering curriculum at an early stage would be advantageous for future development.

5 A look into the future of quantum civil engineering

The concept is to leverage quantum algorithms, qubits and quantum data processing to enhance simulation, optimization and design in civil engineering. A formula frame work is given as:

$$\begin{aligned}&\text{Quantum Civil Engineering}\\&= f\left(\mathrm{Q}_{\text{algorithms}},\ \mathrm{Q}_{\text{hardware}},\ \mathrm{CE}_{\text{structural/fluid/../transportation}},\ \text{Optimization/Uncertainty}\right)\end{aligned} \tag{12}$$

Where *f*(.) defines the coupling of quantum computational models with civil engineering simulation and optimization tasks.

A quantum civil engineering formula based integration framework proposes a structured way to bring the two fields together, Table 3 illustrates some integration. The integration of civil engineering formulas into quantum framework does not mean that physical beams become quantum objects, but rather quantum computation can be used to simulate, optimize or invert complex civil engineering models.

By aligning some well-established engineering formulas with quantum algorithms, this framework would offer conceptual clarity and practical pathways for civil engineers to harness quantum computing. In addition, this would harness the development of domain specific quantum compliers for civil engineering equations.

Table 3 Illustrates civil engineering domains enhanced by quantum computing

Civil engineering domain	Formulas	Algorithms
Structural	Optimization for topology design $min_x f(x) = \frac{1}{2}x^T Kx - F^T$ x	Mapped to QUBO (quadratic unconstrained binary optimization) for quantum annealers: $E(q) = q^T Qq$
Transportation	Optimization for traffic flow $min_x C(x) = \sum_i \sum_j a_{ij} x_i x_j$	Solved with quantum approximate optimization algorithm(QAOA)
Reliability analysis	Probabilistic simulation for load uncertainties: $R = \int_0^\infty P(S < R)\, dS$	Accelerated by quantum Monte Carlo sampling
Geotechnical	Optimization for slope stability $min_x f(x) = W(x)\, s.t F_s \geq 1.5$	Solved with quantum approximate optimization algorithm(QAOA)
Materials	Quantum chemistry simulation to design high strength materials by solving Schrodinger equation: $\widehat{H}\,\lvert\Psi\rangle = E\,\lvert\Psi\rangle$	Implemented via variational quantum eigensolver(VQE)
Fluid	Navier–Stokes equation: $\rho\left(\frac{\delta u}{\delta t} + u \cdot \nabla u\right) = -\nabla p + \mu \nabla^2 u + \rho g$	Integrated with quantum lattice Boltzman method(QLBM) for fluid simulation
Construction	Minimize: $C = \sum_i c_i x_i$ subjected to $\sum_i a_{ij} x_i \leq b_j$ Encode as quadratic unconstrained binary optimization(QUBO):$\mathrm{min} x^T Qx$	Solved using quantum annealing for project scheduling and resource optimization
Environmental	Advection–diffusion-reaction equation: $\frac{\partial C}{\partial t} = D\nabla^2 C - v \cdot \nabla C + R$	Integrated with quantum machine learning(QML) models for environmental forecasting $y = f_{QNN}(x)$
Surveying	Survey station optimization $min_x f(x) = \sum_{i,j} w_{ij} x_i x_j$	Integrated with quantum approximate optimization algorithm(QAOA) for optimal station placement, Variational quantum eigensolver(VQE) for more complex terrain constraints

Quantum civil engineering lies in expressing classical physical systems as linear operators within a quantum state space. A flowchart, Fig. 12, illustrates how quantum computing principles and algorithms flow into civil engineering application, leading to the new field of quantum civil engineering.

By integrating quantum mechanics into civil engineering analysis, design and material science, the framework envisions a new generation of intelligent, adaptive and sustainable infrastructures. Furthermore, quantum-civil engineered approach lays a foundation between quantum data processes, digital twins and AI- driven decision systems which aligns with the next frontier of scientific and technological evolution.

5.1 Application aspects

Quantum Civil Engineering represents a transformative paradigm that integrates quantum computing concepts, technologies, and operations into the design, construction, and management of infrastructure systems. As quantum technologies advance, their applications in civil engineering are expected to be vast and revolutionary [143]. The following highlights key areas where quantum computing could reshape the discipline:

1. Project planning and design play an indispensable role in the successful completion of a project. The influence of quantum algorithm can change method of analysis for site condition, environmental impact and material properties to optimize infrastructural designs for durability, efficiency and sustainability.
2. Multiple analysis and design iterations in structural engineering require excellent precision and efficiency. Quantum algorithms can analyse complicated structural

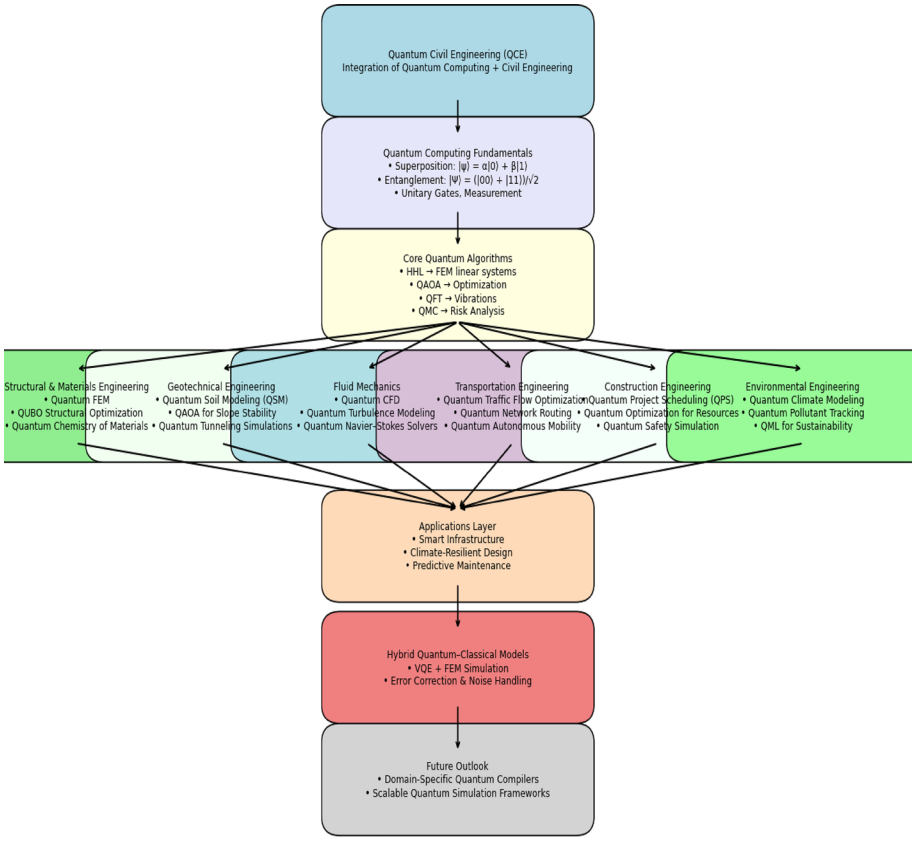

Fig. 12 Flowchart: quantum civil engineering

models, predict potential failure modes and optimize designs to be safe and compliant with codes and specifications.

3. In building sustainability creating eco-friendly alternatives is essential. Simulating molecular reactions at a high precision with quantum computing can design sustainable materials to reduce the use of harmful substances. In addition, energy consumption within buildings can be optimized with quantum algorithm. Light designs and HVAC systems can be optimized to minimize environmental impact and reduce operational cost.
4. Optimizing traffic flow to influence vehicle movement and reduce congestion through modern technology requires improvement. Quantum algorithms can identify optimal routes and minimize delay. Similarly, quantum supply chain optimization would provide means to reduce cost of material procurement, reducing waste and enhancing project efficiency.
5. Quantum computing can accelerate the development of new construction material with superior properties. Simulations through quantum can model material behaviour at molecular and atomic level, which enables innovative material design with enhanced durability and strength.
6. Maintenance strategies can be predicted through quantum computing, which would assist to prevent equipment failure and downtime in construction sites. By analysing machine learning models and sensor data, quantum algorithms can optimize performance of structures and equipment that requires early maintenance.
7. Surveying is a major part of building projects. Quantum sensors would theoretically detect pipeline, underground obstructions and mine shaft before the progress of the impact. Impacting this would save time and funds from unexpected obstacles.
8. Monitoring of air, water and soil pollution with real time data to guide effective environmental protection can be achieved with quantum sensors. In addition, Quantum dot-based sensors can detect pollutant, monitor water quality and other crucial water parameters in real-time. Similarly, quantum computing can optimize irrigation systems to assist farmers and urban cities to make efficient use of water.
9. With quantum computing simulations, modelling the behaviour of complex civil structural systems at a quantum level would attain deeper insights.
10. Image processing techniques are used to process images in order to detect features like cracks. To enhance safety, integrity and longevity of infrastructures, quantum nano sensors can be utilized to monitor structures health.
11. Quantum data analysis and infrastructure monitoring would be a hot topic for civil infrastructures, because data processing would be more accurate to predict complex problems. Through quantum computation more efficient algorithm, improved anomaly detection and predictive analysis would be achieved.

Despite these promising prospects, scaling quantum technologies to practical civil engineering applications remains challenging. Achieving this will require substantial investment, interdisciplinary collaboration, and the development of new standards to guide implementation in the coming decades.

Table 4 Challenges of quantum computing

Challenges category	Specific issue	Impact on civil engineering
Algorithmic	Few domain-specific quantum algorithms	Limits applicability to real-world problems
Computational	High complexity of continuous systems	Difficult to encode and scale
Hardware	Noisy and limited qubits	Reduce accuracy and reliability
Integration	Hybrid models required	Complicates workflows
Educational	Skill gap in quantum literacy	Slow research and adoption

5.2 Challenging but not impossible

Quantum computing has vast potential in civil engineering but significant challenges would definitely be encountered before large scale practical adoptions. Table 4 provided an view on algorithmic complexity and related technical, practical and theoretical issues.

While quantum computing offers solution, in optimizing infrastructure networks, simulating advanced materials and reduces computational cost for large scale models. Current challenges such as algorithmic complexity, qubit limitations and integration barriers would mean that practical quantum applications in civil engineering will be largely theoretical.

6 Conclusions

The progress of the quantum era may seem gradual, yet, like the internet, its transformative potential is undeniable. As quantum computing evolves, its application to civil engineering will enable faster processing of complex and dynamic variables.

Mapping real-world civil engineering challenges to quantum algorithms is essential for advancing efficient and sustainable design. Although still in its early stage, integrating quantum computing into civil engineering promises to enhance efficiency, safety, and sustainability. This review is aimed at physicist, mathematicians, computer scientist and civil engineers to outline state of the art techniques for civil structures. The future of the field will not be shaped solely by steel, concrete, or soil, but also by qubits and quantum algorithms—ushering in the era of Quantum Civil Engineering.

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Author's contributions
IfeOlorun Olofin conceptualized the idea, composed, read and approved the manuscript.

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